**ANALYSIS REPORT ON THE MARKETING CAMPAIGNS ACROSS TIKTOK, FACEBOOK, AND GOOGLE PLATFORMS**

RANTI R. TIMARIO
Data Analyst

## Data Quality and Exploration

- Assign the $ symbol to spend, CPM, and revenue variables.
- Perform remove duplication of data but found none.
- Trim Spaces
- Check the range of dates (January 1 – March 30, 2024)
- Compute the CTR, CPC, CVR, and ROAS using the given marketing data
  Formula used:

| Metric | Formula |
|---|---|
| CTR | $\frac{\text{Clicks}}{\text{Impressions}}$ |
| CPC | $\frac{\text{Total Ad Spend}}{\text{Clicks}}$ |
| CVR | $\frac{\text{Conversions}}{\text{Clicks}}$ |
| ROAS | $\frac{\text{Revenue}}{\text{Ad Spend}}$ |

- Remove the row data with CTR, CPC, CVR, and ROAS result of #VAL or DIV/0.
- Remove ROAS > 10
- Target ROAS should be at least 1.2
- Data After Doing the Data Cleaning and Metric Calculation:
  MarketingData_Processed

## Performance Analysis of Media Platforms Across Key Metrics

A. Click-Through Rate (CTR)

**Table 1.** Significant Difference between the Performance of Various Media Platform on CTR Metric

| Platforms | N | Mean | SD | Comparisons | Mean Diff. | t | $p_t$ | F | $p$ |
|---|---|---|---|---|---|---|---|---|---|
| Facebook | 1064 | 0.571 | 0.208 | FB - Google | -0.529 | -41.608 | <0.001 | 1017.566 | <0.001 |
| Google | 961 | 1.100 | 0.424 | FB - TikTok | -0.063 | -5.066 | <0.001 | | |
| TikTok | 1068 | 0.634 | 0.177 | Google - TikTok | 0.466 | 36.710 | <0.001 | | |

p-value <0.05 = Significant; SD = Standard Deviation

The table above shows the performance of various platforms in terms of Click-Through Rate (CTR). It was shown that Google has the highest mean CTR which is 1.100 and Facebook has the lowest CTR rate (0.571). Using Analysis of Variance with level of significance of 0.05, the results revealed that there is a significant difference between the CTR of various platforms ($F = 1017.566$, $p < 0.05$). With this, Post Hoc Analysis was used specifically the Tukey Test to compare the effectiveness of two platforms in relation to CTR. The results of the data shows that Google Advertising is more effective in attracting customers to click as compared to Facebook ($t=-41.608$, $p < 0.05$). In addition, Google

advertising is more effective than TikTok advertising (t=36.710, p<0.05) while TikTok advertising is more effective than Facebook advertising (t=-5.066, p<0.05). This means that Google ads are the most effective at generating user engagement, making them ideal for driving traffic to websites or landing pages. Since Facebook has the lowest CTR, it may require better targeting or more compelling creatives to remain competitive in capturing user attention.

B. Cost Per Click (CPC)

**Table 2.** Significant Difference between the Performance of Various Media Platform on CPC Metric

| Platforms | N | Mean | SD | Comparisons | Mean Diff. | t | $p_t$ | F | $p$ |
|---|---|---|---|---|---|---|---|---|---|
| Facebook | 1064 | $3.356 | $17.003 | FB - Google | 0.966 | 1.849 | 0.154 | 3.160 | 0.043 |
| Google | 961 | $2.390 | $9.703 | FB - TikTok | 1.215 | 2.389 | 0.045 | | |
| TikTok | 1068 | $2.141 | $5.162 | Google - TikTok | 0.249 | 0.476 | 0.883 | | |

p-value <0.05 = Significant; SD = Standard Deviation

The table above shows the performance of various platforms in terms of Cost Per Click (CPC). It was shown that Facebook has the highest mean CPC which is $3.356 and TikTok has the least CPC ($2.141). This means that it takes $3.356 for every one click of posted campaign or post in Facebook while $2.141 were spent for every one click of posted campaign or post in TikTok. Using Analysis of Variance with level of significance of 0.05, the results revealed that there is a significant difference between the CPC of various platforms (F = 3.160, p < 0.05). With this, Post Hoc Analysis was used specifically the Tukey Test to compare the effectiveness of two platforms in relation to CPC. The results of the data shows that there is no significance difference between the CPC of Facebook Ads and Google Ads (t=1.849, p>0.05). In addition, there is no significant difference between the CPC of Google Ads and TikTok Ads (t=0.476, p>0.05) while there is a significant difference between the CPC of Facebook Ads and TikTok Ads (t=2.389, p<0.05). This implies that the cost of advertising in Facebook is more expensive as compared to TikTok advertising. On the other hand, no statistical evidence to conclude that the cost per click in Google Advertising is more expensive as compared to other platforms. This means that advertising on TikTok is more cost-efficient in generating clicks compared to Facebook. Since Facebook has significantly higher CPC, businesses may need to optimize their targeting or creative strategy on that platform to justify the higher cost and maintain a healthy return on investment.

C. Conversion Rate (CVR)

**Table 3.** Significant Difference between the Performance of Various Media Platform on CVR Metric

| Platforms | N | Mean | SD | Comparisons | Mean Diff. | t | $p_t$ | F | $p$ |
|---|---|---|---|---|---|---|---|---|---|
| Facebook | 1064 | 2.872 | 0.757% | FB - Google | 0.328 | 9.295 | <0.001 | 60.074 | <0.001 |
| Google | 961 | 2.543 | 0.817% | FB - TikTok | 0.330 | 9.604 | <0.001 | | |
| TikTok | 1068 | 2.542 | 0.809% | Google - TikTok | 0.002 | 0.476 | 0.998 | | |

p-value <0.05 = Significant; SD = Standard Deviation

The table above shows the performance of various platforms in terms of Conversion Rate (CVR). It was shown that Facebook has the highest mean percentage

CVR which is 2.872 and TikTok has the least CVR (2.542) although the three platforms are within the target range of 2-4%. Using Analysis of Variance with level of significance of 0.05, the results revealed that there is a significant difference between the CVR of various platforms ($F = 60.074$, $p<0.05$). With this, Post Hoc Analysis was used specifically the Tukey Test to compare the effectiveness of two platforms in relation to CVR. The results of the data showed that there is a significance difference between the CVR of Facebook Ads and Google Ads ($t=9.295$, $p>0.05$). In addition, there is a significant difference between the CVR of Facebook Ads and TikTok Ads ($t=9.604$, $p<0.05$) while there is no significant difference between the CVR of Google Ads and TikTok Ads ($t=0.476$, $p<0.05$). This implies that among the three platforms, Facebook advertising is the most effective channel in turning clicks into conversions when compared to Google and TikTok.

D. Return on Ad Spend (ROAS)

**Table 4.** Significant Difference between the Performance of Various Media Platform on ROAS Metric

| Platforms | N | Mean | SD | Comparisons | Mean Diff. | t | $p_t$ | F | $p$ |
|---|---|---|---|---|---|---|---|---|---|
| Facebook | 1064 | 0.809 | 0.428 | FB - Google | -0.289 | -12.262 | <0.001 | 75.335 | <0.001 |
| Google | 961 | 1.097 | 0.652 | FB - TikTok | -0.148 | -6.471 | <0.001 | | |
| TikTok | 1068 | 0.957 | 0.497 | Google - TikTok | 0.141 | 5.968 | <0.001 | | |

p-value <0.05 = Significant; SD = Standard Deviation

The table above shows the performance of various platforms in terms of Return on Ad Spend (ROAS). It was shown that Google Advertising has the highest ROAS ratio which is 1.097 and Facebook has the lowest ROAS ratio (0.809). This means that Google earns approximately $1.10 for every $1.00 spent on advertising while Facebook and TikTok earns less than $1 for every $1 spent on advertising. Using Analysis of Variance with level of significance of 0.05, the results revealed that there is a significant difference between the ROAS of various platforms ($F = 75.335$, $p<0.05$). With this, Post Hoc Analysis was used specifically the Tukey Test to compare the effectiveness of two platforms in relation to ROAS. The results of the data showed that there is a significance difference between the ROAS ratio of Facebook Ads and Google Ads ($t=-12.262$, $p<0.05$). In addition, there is a significant difference between the ROAS ratio of Facebook Ads and TikTok Ads ($t=-6.471$, $p<0.05$) while there is a significant difference between the ROAS ratio of Google Ads and TikTok Ads ($t=5.968$, $p<0.05$). This implies that among the three platforms, Google advertising is currently the most cost-efficient platform in terms of revenue generation. Although have not achieved the ROAS target of 1.2, Google advertising should be given priority to improved ROAS while ensuring also to improve ad creatives on other platforms to attain the target.

## Performance Analysis of Identified Regions Across Key Metrics

A. Click-Through Rate (CTR)

**Table 5.** Significant Difference between the Regional Performance on CTR Metric

| Regions | N | Mean | SD | Comparisons | Mean Diff. | t | $p_t$ | F | $p$ |
|---|---|---|---|---|---|---|---|---|---|
| West | 784 | 0.757 | 0.365 | W – S | -0.01 | -0.016 | 1.000 | 0.016 | 0.997 |
| South | 791 | 0.758 | 0.366 | W – NE | 0.02 | 0.130 | 0.999 | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Northeast | 778 | 0.755 | 0.368 | W – MW | -0.02 | -0.087 | 1.000 |
| Midwest | 740 | 0.759 | 0.373 | S – NE | 0.03 | 0.147 | 0.999 |
| | | | | S – MW | -0.01 | -0.071 | 1.000 |
| | | | | NE - MW | -0.04 | -0.215 | 0.996 |

p-value <0.05 = Significant; SD = Standard Deviation; W = West, S = South, NE = Northeast, MW = Midwest

The table above shows the performance of identified regions (West, South, Northeast, Midwest) in terms of Click-Through Rate (CTR). It was shown that Midwest region has the highest mean CTR which is 0.759 and Northeast region has the lowest CTR (0.755). Using Analysis of Variance with level of significance of 0.05, the results revealed that there is no significant difference between the CTR on regions (F = 0.016, p>0.05). This implies that the location of the customers as categorize in regions does not specifically affect ad engagement in terms of click-through rate. In other words, regional targeting may not be a key driver of CTR performance**.**

B. Cost Per Click (CPC)

**Table 6.** Significant Difference between the Regional Performance on CPC Metric

| **Regions** | **N** | **Mean** | **SD** | **Comparisons** | **Mean Diff.** | **t** | $p_t$ | **F** | $p$ |
|---|---|---|---|---|---|---|---|---|---|
| West | 784 | $3.254 | $21.609 | W – S | 0.844 | 1.425 | 0.483 | 0.970 | 0.406 |
| South | 791 | $2.410 | $4.843 | W – NE | 0.789 | 1.326 | 0.546 | | |
| Northeast | 778 | $2.465 | $6.087 | W – MW | 0.849 | 1.409 | 0.494 | | |
| Midwest | 740 | $2.405 | $4.299 | S – NE | -0.056 | -0.094 | 1.000 | | |
| | | | | S – MW | 0.005 | 0.008 | 1.000 | | |
| | | | | NE - MW | 0.060 | 0.100 | 1.000 | | |

p-value <0.05 = Significant; SD = Standard Deviation; W = West, S = South, NE = Northeast, MW = Midwest

The table above shows the performance of identified regions (West, South, Northeast, Midwest) in terms of Cost Per Click (CPC). It was shown that West region has the highest mean CPC which is $3.254 and the Midwest region has the lowest CPC ($2.405). Using Analysis of Variance with level of significance of 0.05, the results revealed that there is no significant difference between the CPC on regions (F = 0.970, p>0.05). This implies that geographic location does not have a significant impact on ad cost per click**.** Therefore, regional CPC variations are likely due to random fluctuations or other external variables, and not a consistent pattern.

C. Conversion Rate (CVR)

**Table 7.** Significant Difference between the Regional Performance on CVR Metric

| **Regions** | **N** | **Mean** | **SD** | **Comparisons** | **Mean Diff.** | **t** | $p_t$ | **F** | $p$ |
|---|---|---|---|---|---|---|---|---|---|
| West | 784 | 2.772 | 0.809 | W – S | 0.070 | 1.783 | 0.281 | 75.760 | <0.001 |
| South | 791 | 2.701 | 0.710 | W – NE | -0.070 | -1.776 | 0.285 | | |
| Northeast | 778 | 2.842 | 0.863 | W – MW | 0.483 | 12.057 | <0.001 | | |
| Midwest | 740 | 2.289 | 0.731 | S – NE | -0.140 | -3.560 | 0.002 | | |
| | | | | S – MW | 0.412 | 10.325 | <0.001 | | |
| | | | | NE - MW | 0.553 | 13.784 | <0.001 | | |

p-value <0.05 = Significant; SD = Standard Deviation; W = West, S = South, NE = Northeast, MW = Midwest

The table above shows the performance of identified regions (West, South, Northeast, Midwest) in terms of Conversion Rate (CVR). It was shown that Northeast Region has the highest mean CVR percentage which is 2.842 and the Midwest region

has the lowest CPC (2.289). Using Analysis of Variance with level of significance of 0.05, the results revealed that there is a significant difference between the CVR on regions ($F = 75.760$, $p<0.05$). With this, Post Hoc Analysis was used specifically the Tukey Test to compare the effectiveness of two regions in relation to CVR. The data revealed that CVR is significantly higher in the West than Midwest ($t=12.057$, $p<0.05$), CVR is significantly higher in the South than Midwest ($t=10.325$, $p<0.05$), CVR is significantly higher in the Northeast region than in the Midwest ($t=13.784$, $p<0.05$), and CVR is significantly higher in the Northeast than in South ($t=-3.560$, $p<0.05$). On the other hand, comparisons between west vs south as well as west vs northeast shows no significant difference in CVR ($p>0.05$), suggesting performance in CVR is comparable in those regions. This means that targeting customers in the Northeast and West regions is more effective in driving conversions compared to the Midwest. Since the Midwest shows significantly lower conversion rates, it may be more cost-effective to reduce spend in that region and focus marketing efforts on areas with proven higher conversion performance.

D. Return on Ad Spend (ROAS)

**Table 8.** Significant Difference between the Regional Performance on ROAS Metric

| Regions | N | Mean | SD | Comparisons | Mean Diff. | t | $p_t$ | F | $p$ |
|---|---|---|---|---|---|---|---|---|---|
| West | 784 | 0.999 | 0.568 | W – S | 0.047 | 1.748 | 0.299 | 19.097 | <0.001 |
| South | 791 | 0.952 | 0.493 | W – NE | -0.015 | -0.562 | 0.943 | | |
| Northeast | 778 | 1.015 | 0.595 | W – MW | 0.173 | 6.293 | <0.001 | | |
| Midwest | 740 | 0.826 | 0.484 | S – NE | -0.063 | -2.308 | 0.096 | | |
| | | | | S – MW | 0.126 | 4.584 | <0.001 | | |
| | | | | NE - MW | 0.189 | 6.835 | <0.001 | | |

p-value <0.05 = Significant; SD = Standard Deviation; W = West, S = South, NE = Northeast, MW = Midwest

The table above shows the performance of identified regions in terms of Return on Ad Spend (ROAS). It was shown that the Northeast region has the highest ROAS ratio which is 1.015 and Midwest region has the lowest ROAS ratio (0.826). This means that advertising cost specifically target on Northeast region earns approximately $1.10 for every $1.00 spent while the rest of the regions earns less than $1 for every $1 spent on advertising. Using Analysis of Variance with level of significance of 0.05, the results revealed that there is a significant difference between the ROAS of various platforms ($F = 19.097$, $p<0.05$). With this, Post Hoc Analysis was used specifically the Tukey Test to compare the effectiveness of two regions in relation to ROAS. The results of the data showed that ROAS is significantly higher in West region than in Midwest region ($t=6.293$, $p<0.05$), ROAS is significantly higher in South than in Midwest region ($t=4.584$, $p<0,05$), and ROAS is significantly higher in Northeast region than in Midwest region ($t=6.835$, $p<0.05$). This indicates that marketing efforts in the Northeast, West, and South regions are generating better returns compared to the Midwest. To improve overall efficiency and maximize ROI, budget should be reallocated away from the Midwest toward higher-performing regions like the Northeast, where every advertising dollar brings in more revenue.

## Creative Performance Analysis

A. Correlation Between Video Completion Rate and Conversion

**Table 9.** Significant Relationship between Video Completion Rate and Conversion

| Variables | Pearson-r | p-value | Remarks | Verbal Description |
|---|---|---|---|---|
| Video Completion Rate and Conversion (Purchase) | 0.216 | <0.001 | Significantly Correlated | Low Positive Relationship |

p-value <0.05

Table 9 shows the significant correlation between video completion rate and conversion (purchases). The results shows that video completion rate and conversion are significantly correlated ($r=0.216$, $p<0.05$). Although the correlation is low, it implies that as video completion rate increases, conversion increases and vice versa.

B. Comparison between Creative Types (Video, Image, Carousel, Search, Display)

**Table 10.** Descriptive Statistics of Creative Types Across Key Metrics

| Creative Type | CTR | | CPC | | CVR | | ROAS | |
|---|---|---|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | Mean | SD | Mean | SD |
| Video | 0.632 | 0.178 | 2.214 | 4.495 | 2.645 | 0.790 | 0.983 | 0.477 |
| Image | 0.550 | 0.219 | 3.010 | 1.411 | 2.650 | 0.803 | 0.614 | 0.278 |
| Carousel | 0.539 | 0.211 | 4.604 | 29.209 | 3.003 | 0.773 | 0.747 | 0.453 |
| Search | 1.093 | 0.420 | 2.263 | 9.607 | 2.911 | 0.658 | 1.312 | 0.672 |
| Display | 1.115 | 0.433 | 2.653 | 9.909 | 1.782 | 0.541 | 0.654 | 0.280 |

This table compares different types of advertising creatives - Video, Image, Carousel, Search, and Display across four key performance metrics: Click-Through Rate (CTR), Cost Per Click (CPC), Conversion Rate (CVR), and Return on Ad Spend (ROAS). Display and Search ads have the highest CTR, meaning users are more likely to click on these ad types. However, Search ads also deliver the best ROAS (1.312), indicating the highest return for every dollar spent. Although Video ads have a decent CTR and the second-highest ROAS, their CPC is relatively low, making them a cost-effective option. Image and Carousel ads perform similarly but have lower ROAS, suggesting less return for ad spend. In simple terms, if you're aiming for the best return, Search ads perform best, but Video ads offer a strong balance of engagement and value.

## Product Category and Audience Targeting Performance

A. Product Category Performance Based on ROAS ratio

**Table 11.** Significant Difference between the Performance of Different Products on ROAS Metric

| Products | N | Mean | SD | Comparisons | Mean Diff. | t | $p_t$ | F | $p$ |
|---|---|---|---|---|---|---|---|---|---|
| Protein | 1027 | 1.284 | 0.559 | P – PR | 0.670 | 32.904 | <0.001 | 395.921 | <0.001 |
| Preworkout | 1020 | 0.613 | 0.256 | P – W | 0.536 | 18.977 | <0.001 | | |
| Weightloss | 359 | 0.747 | 0.453 | P - D | 0.229 | 10.061 | <0.001 | | |
| Diet | 687 | 1.055 | 0.533 | PR – W | -0.134 | -4.742 | <0.001 | | |
| | | | | PR – D | -0.442 | -19.423 | <0.001 | | |
| | | | | W – D | -0.308 | -10.252 | <0.001 | | |

p-value <0.05 = Significant; SD = Standard Deviation; P = Protein, PR = Preworkout, W = Weightloss, D = Diet

This table shows how different product categories perform in terms of Return on Ad Spend (ROAS), helping us identify which ones are delivering the most value for our marketing dollars. Protein products perform the best with the highest average ROAS (1.284), followed by Diet products (1.055). In contrast, Preworkout and Weightloss products have much lower ROAS, at 0.613 and 0.747 respectively. The statistical tests using post hoc analysis specifically Tukey Test confirm that these differences are highly significant (all p-values < 0.001). From a business perspective, this suggests we should prioritize ad spend on Protein and Diet products, as they are generating the highest returns, while reconsidering the efficiency of campaigns for Preworkout and Weightloss categories. To improve the ROAS target of 1.2, shift more ad budget toward high-performing products like Protein and Diet, and pause or optimize campaigns for lower-ROAS categories such as Preworkout and Weightloss. Additionally, refine targeting by focusing on high-intent audiences and improve landing page experience to boost conversions without increasing spend.

B. Audience Segment Performance Based on ROAS ratio

**Table 12.** Significant Difference between the Performance of Audience Segments on ROAS Metric

| Audience | N | Mean | SD | Comparisons | Mean Diff. | t | $p_t$ | F | $p$ |
|---|---|---|---|---|---|---|---|---|---|
| Athletes | 1378 | 1.113 | 0.581 | A – FE | 0.453 | 21.931 | <0.001 | 260.031 | <0.001 |
| FitnessEnth | 1028 | 0.660 | 0.338 | A – W | 0.058 | 2.478 | 0.035 | | |
| Weightloss | 687 | 1.055 | 0.533 | FE - W | -0.395 | -15.993 | <0.001 | | |

p-value <0.05 = Significant; SD = Standard Deviation; A = Athlete, FE = FitnessEnth, W = Weightloss

The table above compares the ROAS (Return on Ad Spend) performance across three audience segments: Athletes**,** Fitness Enthusiasts, and Weightloss seekers. The results show that Athletes deliver the highest ROAS (1.113), followed closely by the Weightloss audience (1.055), while Fitness Enthusiasts lag significantly behind (0.660). All differences, especially between Athletes and Fitness Enthusiasts, and between Fitness Enthusiasts and Weightloss, are statistically significant ($p < 0.05$). This implies that targeting Athletes and Weightloss audiences is more profitable and should be prioritized in future campaigns, while advertising to Fitness Enthusiasts may require strategy adjustments or closer examination to improve efficiency. To move closer to the ROAS target of 1.2, shift advertising budget toward Athletes and Weightloss audiences, who already show strong performance. Additionally, conduct targeted message testing and funnel optimization for the Fitness Enthusiasts segment to uncover and address potential conversion barriers.

## Impact of Competitive Events on Key Notes

**Table 13.** Significant Difference between the Impact of Competitive Events on CTR, CPC, CVR, and ROAS Metric

| Metrics | | N | Mean | SD | t | df | p |
|---|---|---|---|---|---|---|---|
| CTR | True | 138 | 0.903 | 0.427 | 4.772 | 3091 | <0.001 |
| | False | 2955 | 0.751 | 0.364 | | | |
| CPC | True | 138 | 2.439 | 0.943 | -0.202 | 3091 | 0.840 |
| | False | 2955 | 2.646 | 12.021 | | | |
| CVR | True | 138 | 2.654 | 0.822 | -0.022 | 3091 | 0.982 |
| | False | 2955 | 2.656 | 0.808 | | | |
| ROAS | True | 138 | 0.841 | 0.477 | -2.308 | 3091 | 0.016 |
| | False | 2955 | 0.955 | 0.544 | | | |

p-value <0.05 = Significant; CTR = Click-Through Rate, CPC = Cost Per Click, CVR = Conversion Rate, ROAS = Return On Ad Spend

Table 13 shows how key ad performance metrics change during competitive events (like holidays when competitor advertising increases). During these periods, Click-Through Rate (CTR) is significantly higher (0.903 vs. 0.751, $p < 0.001$), meaning more people click on ads—likely because more ads are in circulation and consumers are actively browsing. However, Return on Ad Spend (ROAS**)** is significantly lower (0.841 vs. 0.955, $p = 0.016$), indicating that although ads get more attention, they generate less revenue per dollar spent. There are no significant changes in Cost Per Click (CPC) or Conversion Rate (CVR), suggesting that the cost to run ads and the rate of converting clicks to purchases remain stable. This means that ads will get more clicks during competitive times, but they'll be less profitable, so businesses should be strategic with budget allocation and creative differentiation during those high-competition periods.

## Relationship Between Ad Frequency and Conversion Rates

**Table 14.** Significant Relationship between Ad Frequency and Conversion Rates

| Variables | Pearson-r | p-value | Remarks |
|---|---|---|---|
| Ad Frequency and Conversion Rate | <0.001 | 0.989 | Not Significantly Correlated |

p-value <0.05

Table 14 shows that there is no significant relationship between ad frequency and conversion rates (Pearson-r < 0.001, $p = 0.989$). The results reveal that showing ads more frequently does not lead to more conversions. This suggests that increasing how often people see your ads won't necessarily help drive more sales or actions. So, rather than focusing on frequency alone, marketers should prioritize other factors like ad quality, targeting, and timing to improve conversion outcomes.

## RECOMMENDATIONS

### Immediate Optimization Opportunities

To formalize the goal of cutting 30% of spend from the lowest performing areas, there is a need to reconsider the following:

- Facebook has ROAS of 0.809 and CPC of 3.356, so budget can be cut by 30 – 40%.
- Under Creative Types, Image has ROAS of 0.614 Carousel has ROAS of 0.747, which means that they can be cut a budget of around 20 – 30%.
- In terms of Product Categories, Preworkout has ROAS of 0.613 in which marketing may reconsider pausing or look into the campaigns and identify problems that lead to low ROAS ratio such as retargeting or develop new campaigns.
- Fitness Enthusiasts under Audience has a low ROAS of 0.660 which means that budget can be cut by around 40 – 50%.

### Reallocate Budget to High – Performing Areas

Upon identifying the low performing areas, cut budgets can be allocated to high performing areas to produce more revenue and profit margin. The following can be reconsidered after careful analysis of the data:

- In terms of the platform used, Google has a ROAS of 1.097, this means that budget can be increased by 20 – 30% to attain the target ROAS of 1.2.
- In terms of creative types, Search has an ROAS of 1.312 (above the target) and Video has an ROAS of 0.983, so we need to boost Search by adding 25% budget and Video by 20%.
- In terms of product Categories, Protein has ROAS of 1.284 (above the target) and Diet has a ROAS of 1.055, this means that Protein campaigns can be increased by 25% and Diet campaigns can be increased by 15% budget.
- In terms of audience segments, athletes have ROAS of 1.113, and diet has ROAS of 1.055 although below the target but these audience should be prioritized in targeting efforts.

### Bidding Strategy Adjustments Around Competitive Events

During competitive periods, there is a need to reduce 10-15% to counter declining ROAS ratio. There is also a need to focus on retargeting who already showed intent instead of focusing on broad reach.

### Testing Recommendations

Based on the results of the analysis, the following is recommended:

**Test I: Audience Specific Messaging for Fitness Enthusiasts**

| | |
|---|---|
| Hypothesis: | Tailored Messaging fit for Fitness Enthusiasts to increase CVR and ROAS. |
| Method: | A/B Test Control Ad Versus Variant Ad |
| Metrics: | ROAS, CVR |

| | |
|---|---|
| Expected Impact: | Increased 10% CVR and 0.2 in ROAS |
| Timeline: | 2-week test (rollout in 1 week) |

**Test II: Creative Format Optimization**

| | |
|---|---|
| Hypothesis: | Video Ads perform better than Image Ads in Engagement and ROAS |
| Method: | A/B test campaigns using identical messaging in Image vs. Video formats. |
| Metrics: | CTR, ROAS |
| Expected Impact: | Increased 15% CTR and 0.2 in ROAS |
| Timeline: | 3-week test (1 week startup, 2 weeks run) |

**Test III: Landing Page Experiment**

| | |
|---|---|
| Hypothesis: | Dedicated landing pages for each audience segment increase conversion |
| Method: | Split traffic 50/50 between generic and segment-specific pages. |
| Metrics: | CVR, ROAS |
| Expected Impact: | Increased 12% CVR |
| Timeline: | 4-week cycle (1 week build, 3 weeks run) |

**Long Term Strategy**

The table below summarizes the strategy needed to attain the business goals in terms of the following categories:

| | |
|---|---|
| Regional Budget Allocation | • Prioritize Northeast Region (ROAS = 1.015)<br>• Prioritize West Region (ROAS = 0.999)<br>• Deprioritize Midwest Region (ROAS = 0.826)<br>• Allocate future budgets based on ROAS performance by region. |
| Creative Strategy by Platform | • Google Ads: Use Search Ads<br>• TikTok Ads: Prioritize engaging Video Ads since it has low CPC and solid ROAS<br>• Facebook Ads: Cut/Limit Budget, Use only for retargeting or High Performing Segments |
| Audience & Product Category Targeting | • Focus on Athletes and Weight loss Segments<br>• Focus on Protein and Diet Products |
| Quarterly Planning Framework | • Pre-Event (Week 2 – Event): Raise Spend for retargeting, lower for cold traffic<br>• During Event: Use Conservative Bidding; Increased Frequency<br>• Post-Event (Week +1): Reassess performance; Restore Higher Bids where ROAS returns |

## Proposed Timeline

| | |
|---|---|
| Week 1 | Cut low-performing campaigns, reallocate budget |
| Week 2 | Launch A/B Test I |
| Week 3 | Launch A/B Test II |
| Week 4 | Launch A/B Test III |
| Week 5-8 | Monitor Results, optimize, scale what works |

## Risks and Mitigation

| Risk | Mitigation |
|---|---|
| Overspending during low ROAS Events | Use Automated Bidding and Early Event Forecast |
| Under-delivery from Ad Cuts | Gradual Reduction, monitor reach and Frequency |
| Audience Saturation | Retargeting Controls and Exclusion Lists |

## Measurement Plan

Before, during and after the implementation, there should be weekly ROAS tracking by campaign, audience, and platform. In addition, the A/B test performance should be monitored, careful. review of regional and creative type performance and quarterly performance review aligned to the planning cycle.